GROUP 5 · IIT MADRAS DS AND AI LAB · 2026 T3 · MILESTONE 2

SerpentGuard AI — Dataset, SAM Masks & EDA

Offline Indian Snake Segmentation, Classification & Safety Decision Support (87 / 87 Automated Checks Passed)

PROCESSED DATASET	TAXONOMY & TIERS	WHO MEDICAL PRIORITY	SAM MASKS & INTEGRITY
121,566	**184 Classes**	**16 WHO Classes**	**82,354 Masks**
VERIFIED SNAKE PHOTOS	**56 TIER-1 · 128 TIER-2**	**6 CAT 1 · 10 CAT 2 CLASSES**	**ON 85,067 TRAIN PHOTOS**
62,772 sightings by 15,842 photographers	+ 97 Unknown Pool species (815 photos)	15,719 photos (13.0% of class photos)	0 shared observers/hashes across splits

Three Attached Kaggle Datasets & Presentation Scope

I. DATASET ARCHITECTURE & LABELS

- `serpentguard-indian-snakes-clean` (121,566)
- `serpentguard-m2-splits` & Cleaning Funnel
- Long-Tail Imbalance (Gini 0.73) & 14 Families
- WHO SEARO 2016 Category 1 & 2 Coverage

II. SPLITS, COVARIATE SHIFTS & PIXEL AUDIT

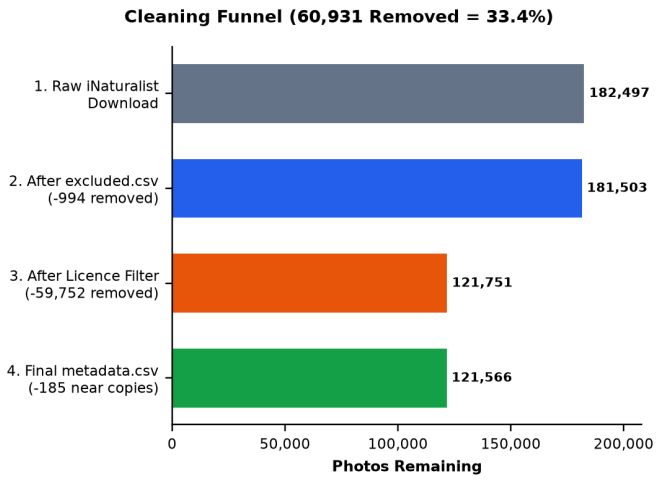
- Observer-Grouped Splits & 2022+ India Test Rule
- Hidden Shifts: Portrait Lean & Country Shortcut
- Snake Box Size at 224/320/384 px & Non-JPG Outliers
- 6,249-Photo Pixel Quality & ImageNet Norms

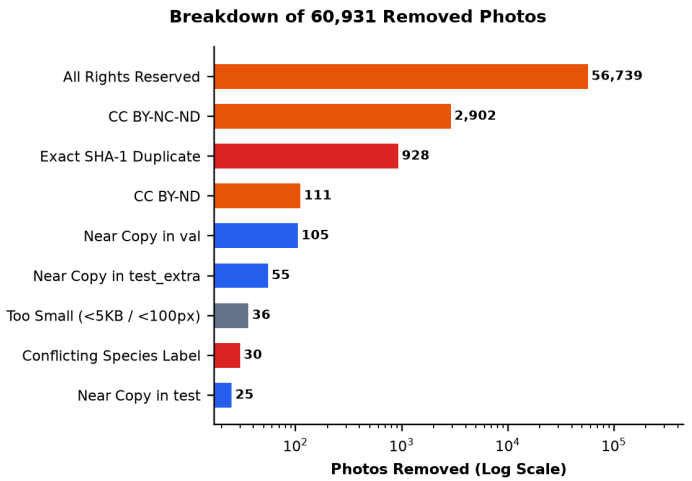
III. EMBEDDINGS, LOOKALIKES & SAM MASKS

- DINOv2 Embeddings (Domain AUC 0.50–0.56 & Cleanlab)
- 9 M1 Lookalike Pairs + Unlisted Venomous Relatives
- `serpentguard-sam-masks` (82,354 Masks Audit)
- Geography, Licences & Milestone 3 Action Plan

Group 5: Jayesh Bansal (Team Lead) · Vikas · Puneet Bajaj · Manas Rastogi

From 182,497 Raw Downloads to 121,566 Verified Photos Across 3 Kaggle Datasets



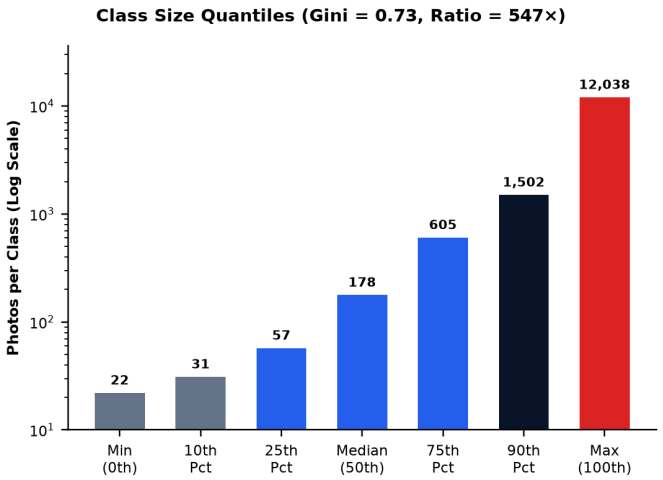


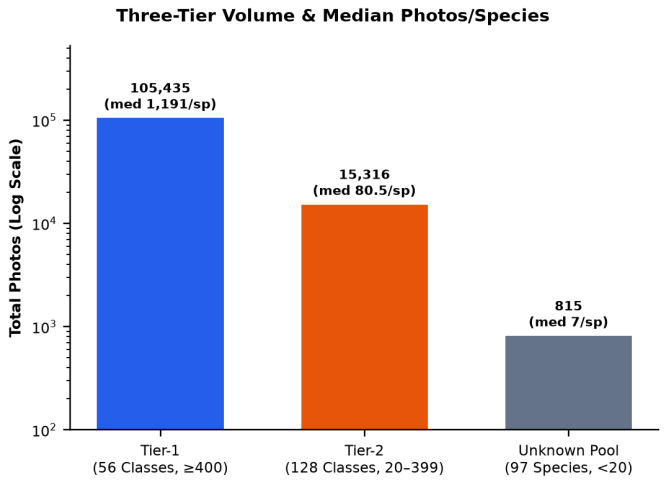
Three-Stage Cleaning & Disk Verification

- **Stage 1 Hygiene (`excluded.csv` = 994 rows)**: Removed **928** exact SHA-1 duplicates, **30** conflicting-species uploads, and **36** broken/tiny files (<5 KB: 19; <100 px: 17) → **181,503** rows in `splits.csv`.
- **Stage 2 Licence Filter (-59,752 rows)**: Removed **56,739** all-rights-reserved and **3,013** no-derivatives photos → **121,751** rows.
- **Stage 3 Split Deduplication (-185 rows)**: Dropped near copies within 4 pHash bits from `val` (-105), `test_extra` (-55), and `test` (-25) without touching `train` → **121,566** final rows.
- **100% Disk & Schema Match**: Every row in `metadata.csv` (33 raw columns) matches an image on disk; only `latitude/longitude` have **763** gaps (0.63%).

KAGGLE DATASET HANDLE	CORE FILES & DIMENSIONS	VERIFIED INTEGRITY CHECK
`serpentguard-indian-snakes-clean`	`metadata.csv` (121,566 × 33) · `classes.csv` (184 × 13) · `processing_report.json`	0 missing files on disk · 0 byte mismatches (1,000 sampled) · 0 SHA-1 errors
`serpentguard-m2-splits` & `sam-masks`	`splits.csv` (181,503 × 31) · `excluded.csv` (994 × 25) · `mask_stats.csv` (85,067 clean train rows)	100% of `train` photos (85,067) linked to a precomputed SAM record; 82,354 have a mask

Severe Long-Tail Skew (Gini = 0.73, Ratio = 547.2×): 13 Classes Hold 50% of Photos



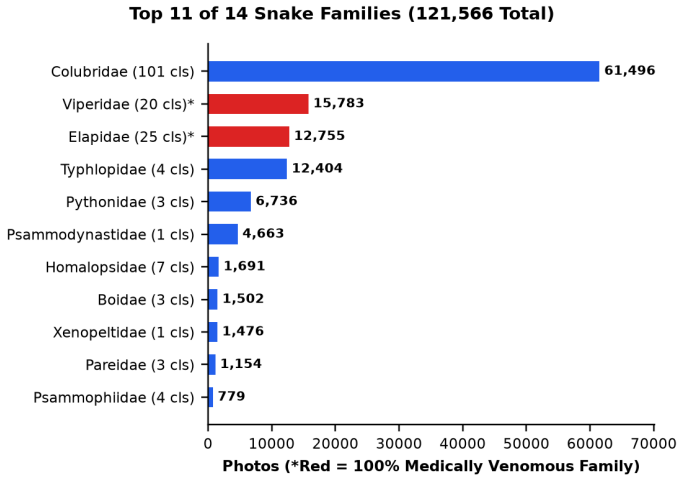


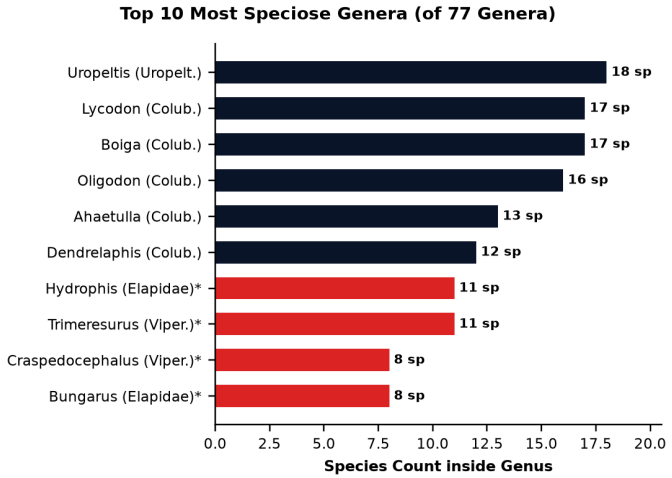
Quantifying the Imbalance

- **Effective Classes = 32.6 (vs. 184 Real)**: Top 10 classes hold **45.1%** of all class photos (120,751), while the 50 smallest classes hold just **1.6%** (73 classes have <100 photos).
- **Three-Tier Split**: **Tier-1** (56 classes, ≥400 photos) holds **105,435 photos** (median 1,191/sp); **Tier-2** (128 classes, 20–399) holds **15,316 photos** (median 80.5/sp).
- **Open-Set Holdout & M1 Core-27**: **97 rare species** (≤19 photos; 815 total) form `unknown_pool`; all **27 M1 Core species** sit in Tier-1 (**51,556 photos** = 42.7%).

IMBALANCE METRIC	ALL SPLITS (184 CLASSES)	TRAIN ONLY (85,067 PHOTOS)	MODELING IMPLICATION
Largest / Median / Smallest Class	12,038 / 177.5 / 22 photos	8,760 / 120.5 / 10 photos	Plain accuracy reflects only the top ~13 classes; must track macro-F1 & per-tier recall
Imbalance Ratio & Effective Classes	**547.2×** · Gini **0.73** · N_{eff} = 32.58	**876.0×** · Gini **0.73** · N_{eff} = 31.88	Caps + tempered sampling (1/√N) in notebook 06 bring train to 15.6× and 118.6 effective classes; M3 compares losses

Colubridae Holds 50.6% of Photos; Viperidae & Elapidae Form 28,538 Venomous Photos



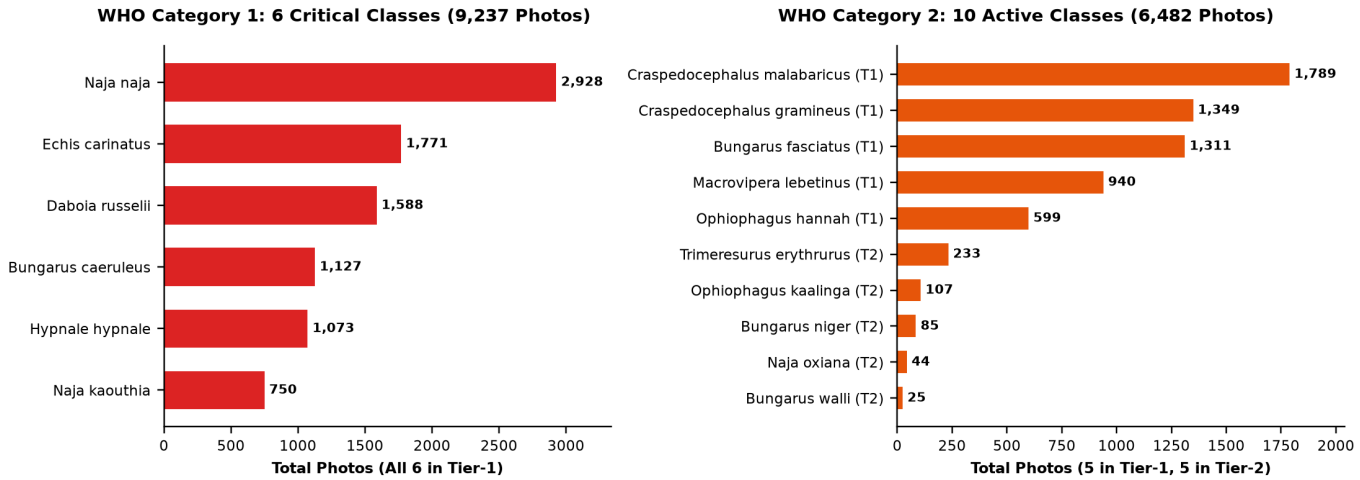


Clade Structure & Subspecies

- **Family Concentration**: **Colubridae** dominates with **61,496 photos** (145 species, 101 classes, 33 genera). At the tail, *Xenodermidae* (3 photos, 1 sp) sits entirely in `unknown_pool`.
- **100% Venomous Families (23.5%)**: **Viperidae** (15,783 photos, 20 classes) and **Elapidae** (12,755 photos, 25 classes) are medically venomous across every species.
- **Subspecies & Synonyms (5.0%)**: **6,099 photos** carry a subspecies/synonym in `record_taxon` (e.g. *C. helena helena* 649, *E. carinatus sochureki* 633) folded into the 281 species.

TAXONOMIC LEVEL	COVERAGE IN DATASET	WHY IT MATTERS FOR MULTI-TASK CLASSIFICATION
14 Families	Colubridae (61.5k), Viperidae (15.8k), Elapidae (12.8k), Typhlopidae (12.4k)	Family head separates 100% venomous Elapids/Vipers from Typhlopids/Colubrids
77 Genera (30 Monotypic)	*Uropeltis* (18 sp, 434), *Lycodon* (17 sp, 3,819), *Boiga* (17 sp, 3,577), *Oligodon* (16 sp, 2,416)	Coarse genus prediction can give a fallback for rare 20-photo Tier-2 species; M3 measures whether it helps

16 Active WHO Classes (15,719 Photos): All 6 Category 1 Species Have 750+ Photos

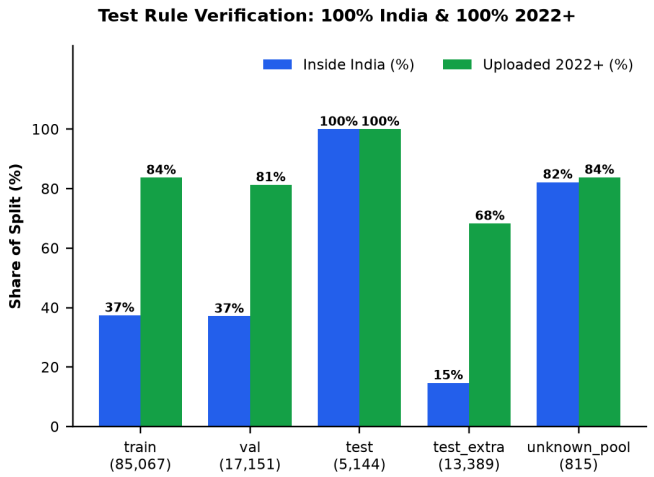


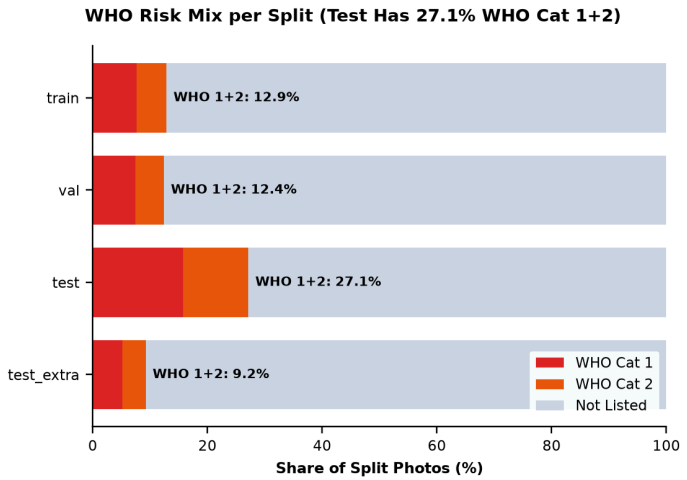
Medical Priority Breakdown

- **WHO Category 1 (6 Classes — 9,237 Photos, 7.6%)**: Highest mortality/morbidity. Median **1,357.5 photos/class**; all 6 sit in Tier-1 (*Naja naja* 2,928 down to *Naja kaouthia* 750).
- **WHO Category 2 (12 Species, 10 Classes — 6,501 Photos, 5.3%)**: Median **416 photos/class**; 5 in Tier-1, 5 in Tier-2, plus 2 rare species (19 photos) in `unknown_pool`.
- **Not Listed (263 Species, 168 Classes — 105,828 Photos, 87.1%)**: Median **153.5 photos/class**; "not listed" never means harmless.

WHO CATEGORY	SPECIES / ACTIVE CLASSES	PHOTOS (SHARE)	MEDIAN PHOTOS / CLASS	TEST SPLIT COVERAGE WARNING
Category 1 (Critical)	6 species · 6 classes (all Tier-1)	9,237 (7.6%)	**1,357.5**	All 6 classes have 22–324 India 2022+ test photos
Category 2 (Important)	12 species · 10 classes (5 T1, 5 T2)	6,501 (5.3%)	**416.0**	**3 classes have 0 test photos** (*M. lebetinus, N. oxiana, B. walli*) → evaluate on `test_extra`

Zero Shared Observers/Hashes Across Splits & 100% India 2022+ Test Set



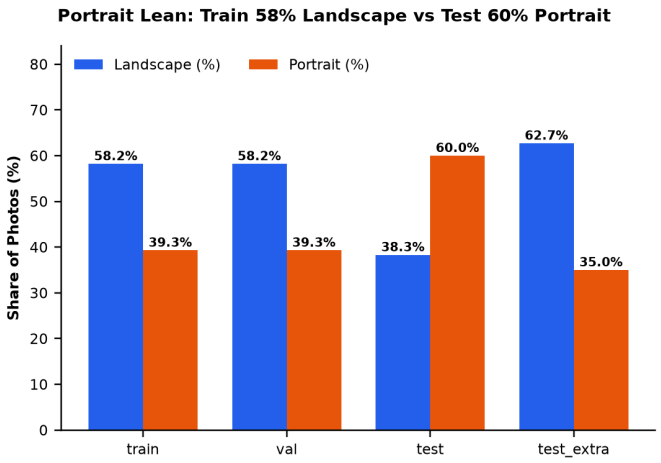


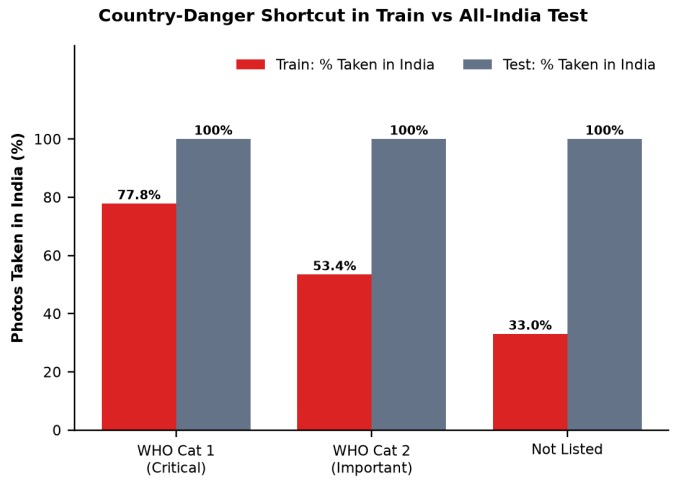
Strict Grouping & Test Rule

- **Zero Leakage Across Train, Val & Test**: Verified **0 shared** `observer_id`, `obs_id`, `sha1`, and exact `phash` across train, val and the held-out fold (69.2% of photos belong to multi-photo sightings). `test` and `test_extra` come from that one fold and share 73 photographers.
- **100% India & 2022+ Test Rule (0 Violations)**: All **5,144 `test` photos** were taken in India and uploaded ≥2022-01-01 (vs. ~37% India in `train`/`val`).
- **36 Classes Have 0 Test Photos**: Due to the strict India 2022+ filter, 36 of 184 classes (including 3 WHO Cat 2 species) have `test = 0`: 33 are scored on `test_extra`, 3 have neither.

SPLIT PARTITION	PHOTOS (SHARE)	IN INDIA %	UPLOADED 2022+ %	WHO CAT 1 + CAT 2 SHARE
`train` · `val`	85,067 (70.4%) · 17,151 (14.2%)	37.4% · 37.1%	83.7% · 81.3%	12.88% (7.60% + 5.27%) · 12.42% (7.38% + 5.04%)
`test` · `test_extra`	5,144 (4.3%) · 13,389 (11.1%)	**100.0%** · 14.7%	**100.0%** · 68.3%	**27.14%** (15.77% + 11.37%) · 9.24% (5.16% + 4.08%)

Three Hidden Shifts: Portrait Lean in Test, Country Shortcut & Thin Test Observers



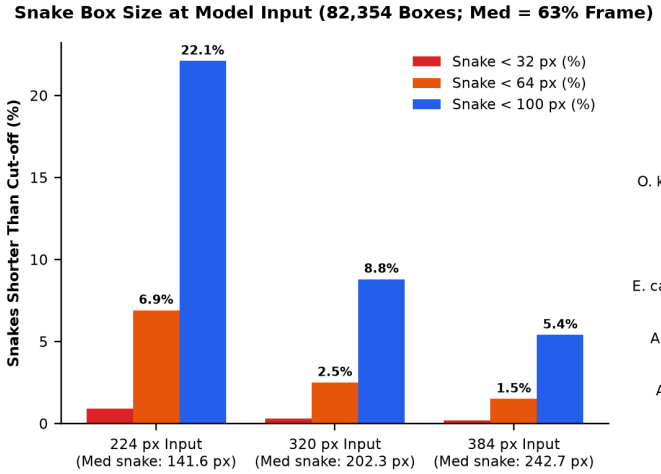


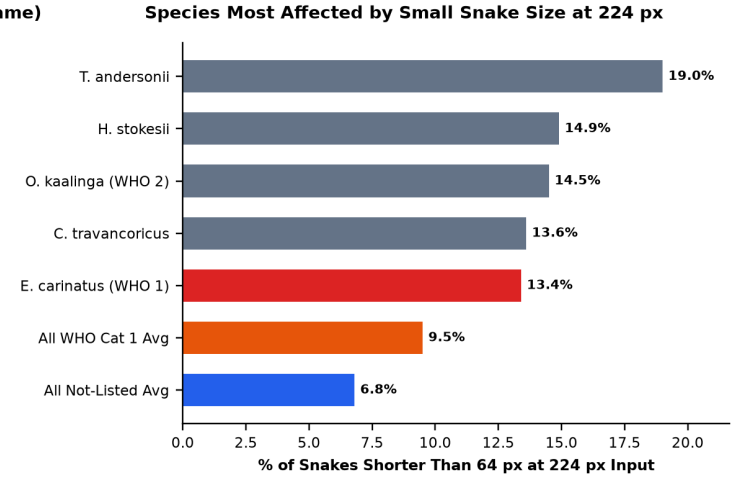
Why Deeper Checks Matter

- **Portrait Lean**: `train` is **58.2% landscape**, whereas `test` is **60.0% portrait**. One photographer took **28.1%** of `test` (98.1% portrait); without them it is 45.0% → aspect-preserving padding and scores both ways.
- **Country-Danger Shortcut in Train**: In `train`, **77.8% of WHO Cat 1** photos are from India vs. only **33.0% of Not Listed**. Background scenery could become a false hint on 100%-India `test`; M3 checks this.
- **Near-Copies Removed & Thin Test Observers**: **0** val/test photos sit within Hamming ≤4 bits of train (185 removed); **71 of 148** test classes have ≤3 test photographers (vs. only 7 of 184 in train ≤5).

HIDDEN AUDIT DIMENSION	MEASURED FINDING IN NOTEBOOK	MITIGATION FOR MILESTONE 3
pHash Near-Copies (≤4 bits)	185 removed (105 `val`, 25 `test`, 55 `test_extra`); 0 left within ≤4 bits; 111 / 27 / 63 within ≤8 bits	Done in notebook 03; a control copy with 3 flipped bits is caught, one with 6 is not (5–8 bits are mostly unrelated photos)
Test Photographer Concentration	71 of 148 test classes have ≤3 photographers (13 classes have 1 photo from 1 person)	Compute test confidence intervals via photographer-grouped bootstrap; also report scores without the top photographer

At 224 px Input, 6.9% of Snakes Are <64 px Long (13.4% for Echis carinatus)



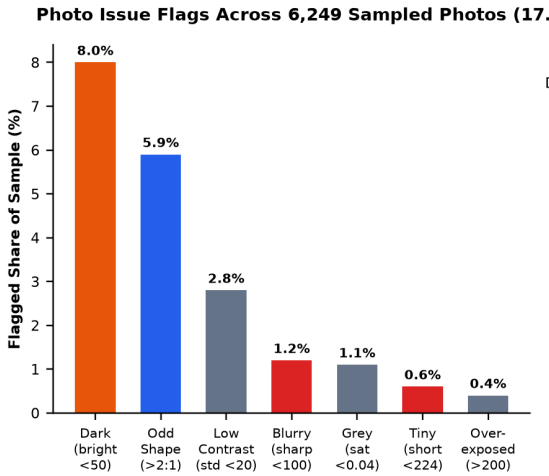


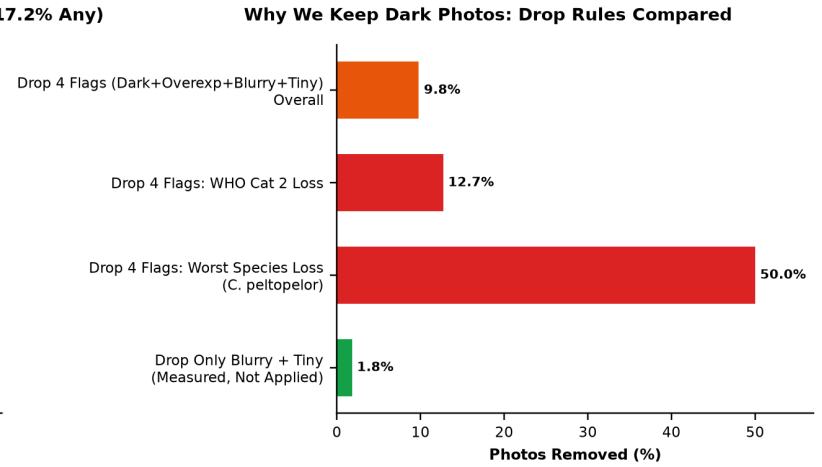
Resolution & Foreground Scale

- **500 px Medium Standard (16.4 GB Total)**: **99.63%** of photos have a long side of exactly 500 px (median 500×375 px, 127 KB). Only **0.70%** have short side <224 px.
- **Snake Box Length (82,354 Train Boxes)**: Median snake spans **63% of the photo's long side** (141.6 px at 224 input). However, **9.5% of WHO Cat 1 snakes** shrink below 64 px at 224 px (led by *Echis carinatus* at **13.4%**).
- **Larger Input Helps Small Snakes**: <64 px snakes fall from **6.9%** at 224 px to **2.5%** at 320 and **1.5%** at 384. We start at 320 px; M3 compares input sizes and masked crops.

FILE & HOST AUDIT	QUANTIFIED FINDING	DATALOADER ENGINEERING RULE
407 Non-JPG URLs & 26.2 MB Outliers	Top 100 bytes/pixel outliers are 64 PNGs & 32 GIFs saved with `.jpg` extension (max 26.2 MB)	Always call `.convert("RGB")` (fixes 14 PNGs, 8 with alpha, found in 6,249 file audit; 0 EXIF rotations)
Two Hosts = Licence Split	`static.inaturalist.org` was 31.3% of 181,503 rows, all but 4 all rights reserved; kept set is 99.997% S3 open-data bucket	Licence filter did not shift file size: kept vs. removed median 127 vs 130 KB

100% pHash Integrity, 0.1% Faces & Why We Keep Dark Nocturnal Snake Photos



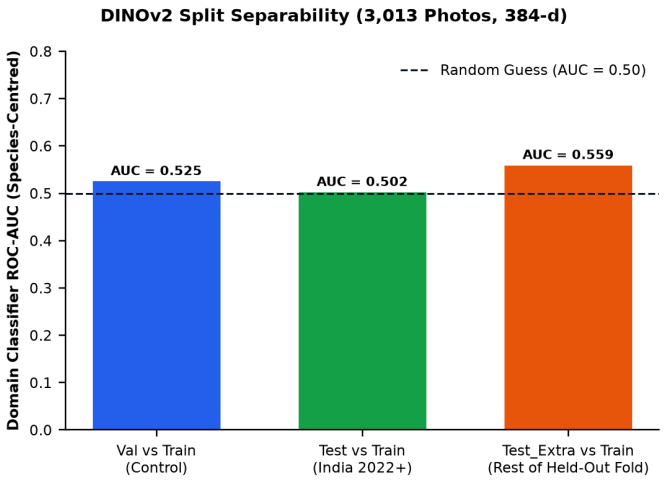


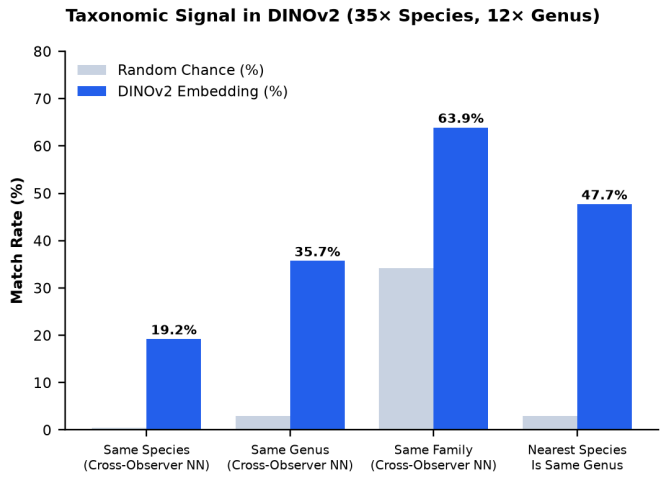
Pixel Quality & Cleaning Rule

- **6,249 Photos Measured (234 s, 0 Read Errors)**: **100.0%** match stored pHash (Hamming 0) and metadata dimensions; YuNet detects human faces in just **0.1%** of photos.
- **Why Not Drop Dark Photos (8.0%)?** Dropping dark + overexposed + blurry + tiny removes **9.8%** overall, hits **WHO Cat 2 hardest (12.7%)**, and removes **50%** of one species (*C. peltopelor*). Exposure correction handles them instead.
- **Keep ImageNet Normalisation**: Our training set is only slightly darker than ImageNet defaults by `[−0.035, −0.024, −0.040]`, and the largest split gap is `0.104` SD (test, blue).

CHANNEL / QUALITY METRIC	TRAIN SAMPLE (N = 3,893)	IMAGENET DEFAULT	PREPROCESSING ACTION
RGB Mean & Std	Mean `[0.450, 0.432, 0.366]` · Std `[0.237, 0.224, 0.224]`	Mean `[0.485, 0.456, 0.406]` · Std `[0.229, 0.224, 0.225]`	Normalise with ImageNet stats for pretrained encoders + keep brightness/contrast jitter
Cleaning Rule	Dropping only **blurry** (<100 Laplacian) + **tiny** (<224 px) would remove **1.8%**; we drop neither	Preserves nocturnal kraits, wolf snakes & pit vipers	CLAHE for dark photos; letterbox odd aspect ratios (>2:1 = 5.9% overall, 14.1% in `test`)

No Strong Domain Shift Found (Test AUC = 0.502) & Strong Clade Signal (12× Genus Match)



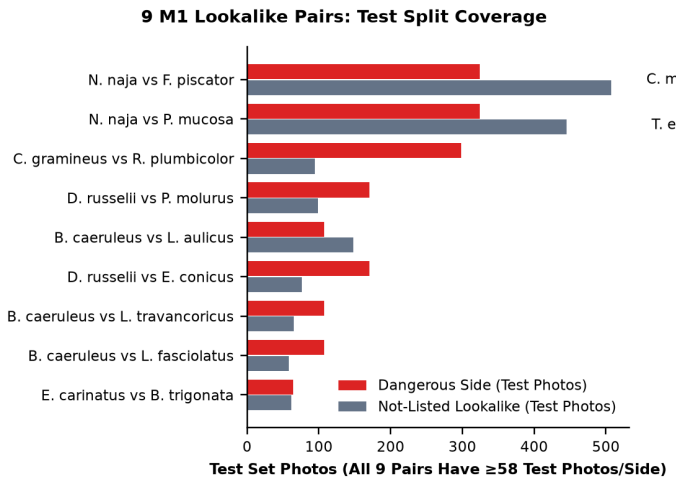


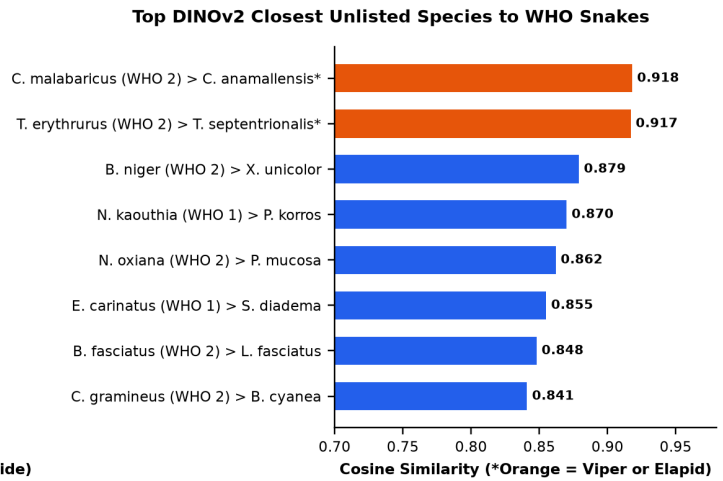
What DINOv2 Reveals

- **Splits Look Visually Close**: A species-centred, observer-grouped classifier separating `train` from `val`, `test`, and `test_extra` gets **AUC = 0.525, 0.502, and 0.559** (≈0.50 random guess). This doesn't prove there is no shift.
- **Test Coverage Matches Val**: Median cosine similarity to the closest same-species training photo is **0.552 in `test`** vs. **0.547 in `val`** and **0.570 in `test_extra`**.
- **Strong Hierarchical Signal**: Cross-photographer nearest neighbours match the **same species 19.2%** (35× chance), **same genus 35.7%** (12× chance), and **same family 63.9%**.

EMBEDDING AUDIT TASK	QUANTIFIED RESULT	ACTIONABLE FOLLOW-UP
Cross-Observer Near-Duplicates	9 photos at cosine ≥0.90; **0 at ≥0.95** in this 3,013-photo sample	Inspect the 9 high-similarity cross-observer photos before locking final test evaluation
Cleanlab Confident Learning	Ranked `label_issue_candidates.csv` (classifier 25.9%): 17.7% cross WHO categories; 9 match known lookalike pairs	Top flags are tiny/night/occluded snakes rather than wrong labels → hand-verify cross-WHO & lookalike flags only

All 9 M1 Lookalike Pairs Have ≥58 Test Photos; Embeddings Reveal Unlisted Relatives



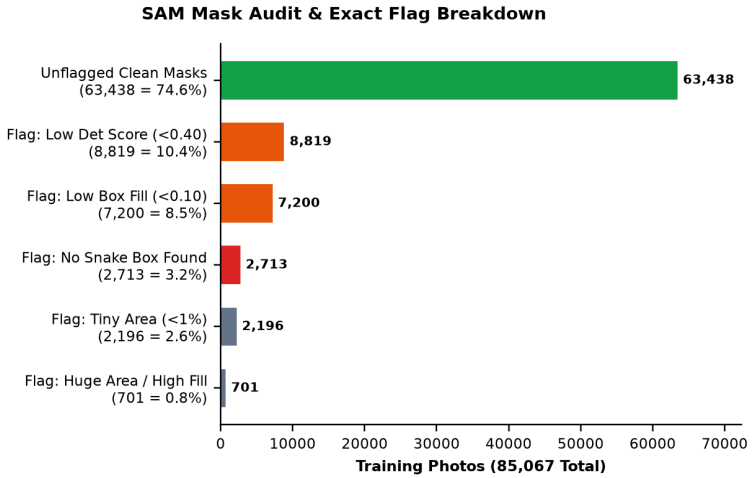


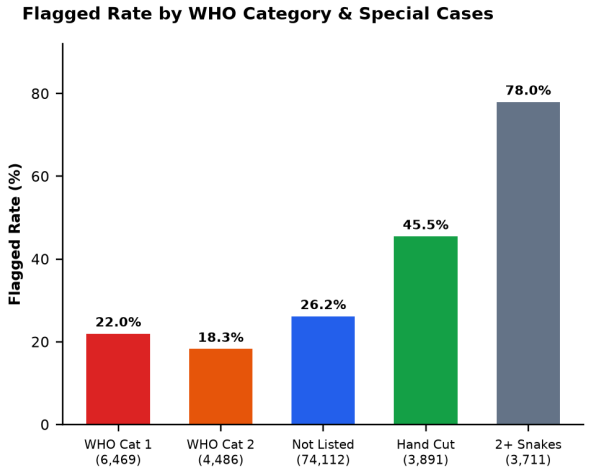
Beyond the 9 M1 Pairs

- **All 9 M1 Pairs Are Test-Ready**: Every pair has ≥58 test photos per side; in DINOv2 space, **4 of 9** dangerous snakes rank their named lookalike within the **top 5 closest species of 183**.
- **Sharpness Gap Warning**: *Echis carinatus* photos are sharper than *Boiga trigonata* (median 3,012 vs 1,747, p = 0.030, 30 photos per side) → blur augmentation is in notebook 06 so models don't cheat via sharpness.
- **Critical Discovery ("Not Listed" ≠ Harmless)**: For 15 of 16 WHO species, the closest "not_listed" species in DINOv2 isn't on our pair list; **5 of 16 are vipers or elapids** (e.g. *C. anamallensis* 0.918, *T. septentrionalis* 0.917).

LOOKALIKE CATEGORY	REPRESENTATIVE PAIRS & COSINE SIMILARITY	SAFETY UI & EVALUATION RULE
Dangerous vs. Not Listed (9 Pairs)	*B. caeruleus* vs *L. aulicus* (0.767, rank 7) · *E. carinatus* vs *B. trigonata* (0.753, rank 3) · *D. russelii* vs *P. molurus* (0.730, rank 8)	Track pair-wise false-negative rate (calling WHO Cat 1/2 not listed) with 10× asymmetric loss penalty
Venomous vs. Unlisted Congener	*C. malabaricus* → *C. anamallensis* (0.918) · *T. erythrurus* → *T. septentrionalis* (0.917) · *B. caeruleus* → *B. bungaroides* (0.806)	Never label "WHO Not Listed" as "Non-Venomous"; use family/genus heads to flag all Elapids & Vipers as venomous

82,354 Masks on 85,067 Train Photos: 74.6% Clean Unflagged & 3,891 Hand Cutouts



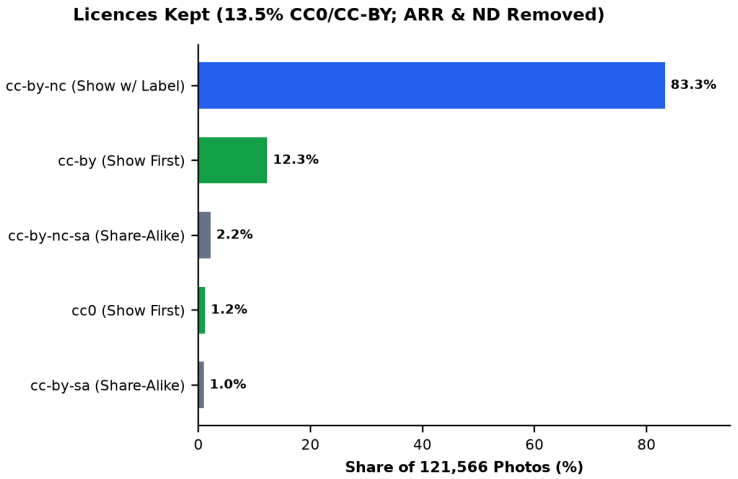


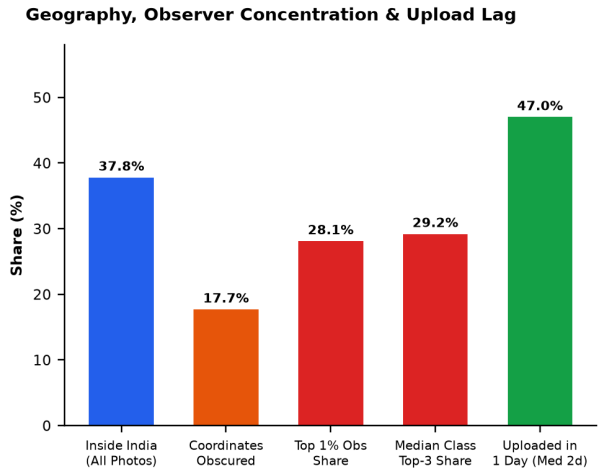
SAM Mask Quality & Thresholds

- **Complete Training Coverage**: `mask_stats.csv` covers **100.0% of `train`** (85,067 photos across all 184 classes: 6,469 WHO Cat 1; 4,486 WHO Cat 2; 74,112 Not Listed).
- **Exact Flagging Rule (100% Verified)**: **63,438 masks (74.6%)** pass all quality gates (`det_score`: 0.40–0.971, `mask_area_fraction`: 0.01–0.60, `fill_ratio`: 0.10–0.95).
- **Hand Removal & Multi-Snake Handling**: Detector automatically cut out human hands/gloves in **3,891 photos (4.6%)**; **2,713 photos (3.2%)** had no snake box (33.1% of them thin *Ahaetulla* vine snakes).

SAM MASK SUBSET	PHOTO COUNT (SHARE)	HOW WE USE IT IN MILESTONE 3 TRAINING
Unflagged Clean Masks	**63,438 (74.6%)** · WHO Cat 1: 78.0% clean, Cat 2: 81.7% clean	Train segmentation head strictly on unflagged masks (`mask_valid`); whole photo is the default input, M3 compares masked modes
Flagged Masks (21,629 = 25.4%)	Low score (8,819) · Low fill (7,200) · No snake (2,713) · Tiny area (2,196) · Huge/High fill (701)	Fallback to full-frame classification without mask suppression so thin arboreal snakes aren't dropped

37.8% India Photos, Mid-Year Lean, Observer Gini = 0.73 & Licence-Clean Dataset





Real-World Context & Governance

- **Geography & Seasonality**: **37.8%** of all photos are inside India (Western Ghats, NE & urban hubs) and **17.7%** have obscured coordinates. July–October holds 46.4% of Indian photos vs. 41.4% elsewhere, a mild **mid-year lean**.
- **Observer Concentration (15,842 Photographers)**: Gini = **0.73**; the top 1% of observers took **28.1%** of all photos (max 2,189 by one user). In 10 small classes, top-3 observers supply **80%–100%** of training photos.
- **Licence Filter**: All-rights-reserved and no-derivatives photos are removed. Of the rest, **13.5%** are `CC0/CC–BY` (13 classes have 0), **83.3%** are `CC–BY–NC`, and **3.1%** are share-alike.

GOVERNANCE / CONTEXT DIMENSION	KEY NUMBER FROM NOTEBOOK	OPERATIONAL RULE FOR SERPENTGUARD AI
Observer Concentration per Class	Median 29.2% of a class's train photos come from its top-3 photographers	Notebook 06 caps training at 50 photos per observer per class and 1,000 per class, with camera-style augmentation
Licence & Redistribution Policy	13.5% Show-First (CC0/CC-BY) · 83.3% Non-Commercial · 3.1% Share-Alike	Datasets stay private; restricted photos stay out of the dataset, repo and docs; credit photographers in reports

What Is Completed in Milestone 2 vs. Next in Milestone 3 & Group 5 Team

PIPELINE STAGE	STATUS	CONCRETE ENGINEERING ACTION DERIVED FROM EDA
1–4. Hygiene, Licences & Dedup	Done in M2	33-col metadata + licence filter (59,752 removed), 500 px JPEG standard, 994 excluded + 185 cross-split near copies removed (**87/87 checks PASS**)
5–8. Splits, Labels & Train Norms	Done in M2	Observer-grouped splits (0 leakage), Cleanlab ranking, ImageNet norms kept (train `[0.450, 0.432, 0.366]` is close); 0 near-copies within ≤4 bits
9. Mask Modes & Input Size	Built in M2	Notebook 06: letterbox to 320×320, whole photo by default, **63,438 usable SAM masks** as target (`mask_valid`); M3 compares masked modes and 224 / 320 / 384 px
10. Tail & Style Augmentation	Built in M2	Train-only augmentation by tier, camera-style transforms, blur and exposure correction; caps + 1/√N sampler with WHO boost (1.60, 1.35)
11. Multi-Head & Safety-Aware Loss	Next · M3	Joint **Species (184) + Genus (69) + Family (13) + WHO Risk** heads, class-balanced vs. focal loss, asymmetric WHO recall & `unknown_pool` abstention

Group 5 Project Team — IIT Madras BS in Data Science & AI Lab (2026 T3)

JAYESH BANSAL · TEAM LEAD	VIKAS · TEAM MEMBER	PUNEET BAJAJ · TEAM MEMBER	MANAS RASTOGI · TEAM MEMBER
Team lead: coordination and distributing the work	Preprocessing pipeline (notebook 06), presentation, help with the analysis	Data collection, cleaning, splits, licence filter, Kaggle datasets, mask pipeline, notebooks 01 to 05 and 07, fixes to 06, 08 and 09	Full exploratory analysis, with help from Vikas; extra-source research and first versions of notebooks 08 and 09